Final Report: NaPMT3 Function-Inference Stress Test — GO:0004766 (Spermidine Synthase Activity)

Executive Judgment

Verdict: Wrong subfamily (mis-placed) — Failure Mode 1 (Granularity / family-vs-subfamily error)

The seed hypothesis — that NaPMT3 (A0A314LG79, *Nicotiana attenuata*) has spermidine synthase activity (GO:0004766) — is **refuted**. NaPMT3 is a **putrescine N-methyltransferase** (PMT; EC 2.1.1.53; GO:0030750), not a spermidine synthase (SPDS; EC 2.5.1.16; GO:0004766). The TreeGrafter annotation is a classic family-vs-subfamily granularity error: the GO term was propagated from the PANTHER family-level node PTHR11558 ("SPERMIDINE/SPERMINE SYNTHASE"), which encompasses both true aminopropyltransferases and the functionally diverged PMT subfamily. NaPMT3 is correctly assigned to subfamily PTHR11558:SF53 ("PUTRESCINE N-METHYLTRANSFERASE 1"), but the GO term was inherited from the broader family rather than the subfamily.

The single most decisive piece of evidence is that NaPMT3 is **99.7% identical** (387/388 residues) to the experimentally characterized NaPMT1 (Q93XQ5), which has been directly demonstrated to function as a putrescine N-methyltransferase in nicotine biosynthesis, and shares only 50–72% identity with bona fide spermidine synthases. The two enzyme activities — SAM-dependent methyl transfer (PMT) versus dcSAM-dependent aminopropyl transfer (SPDS) — reside on entirely different branches of the GO transferase hierarchy.

Summary

NaPMT3 from *Nicotiana attenuata* was annotated with GO:0004766 (spermidine synthase activity) by the TreeGrafter/PANTHER automated phylogenetic pipeline. This annotation is incorrect. Through sequence analysis, domain architecture comparison, active-site residue

alignment, GO hierarchy mapping, and literature review, we demonstrate that NaPMT3 is a putrescine N-methyltransferase (GO:0030750) — the first committed enzyme in nicotine alkaloid biosynthesis. The protein catalyzes the SAM-dependent methylation of putrescine to form N-methylputrescine, a fundamentally different reaction from the decarboxylated SAM (dcSAM)-dependent aminopropyl transfer catalyzed by spermidine synthases.

The mis-annotation arises because PMTs evolved from spermidine synthases and retain the same protein fold (PABS domain, Pfam PF01564). Both enzyme families belong to the PANTHER family PTHR11558, but they are functionally distinct subfamilies. The TreeGrafter pipeline propagated the GO term from the family-level ancestral node rather than respecting the subfamily-level assignment (SF53, "PUTRESCINE N-METHYLTRANSFERASE 1"). This error systematically affects all PMT proteins grafted into this PANTHER family, including the experimentally characterized NaPMT1 (Q93XQ5), which also carries the erroneous GO:0004766 annotation via IEA.

NaPMT3 differs from the experimentally validated NaPMT1 by a single amino acid substitution (S369F) located outside the catalytic PABS domain, with 100% identity across all 12 annotated active-site residues and the catalytic aspartate (D255). This near-identity provides overwhelming evidence that NaPMT3 shares NaPMT1's PMT activity. The correct curation action is to **replace GO:0004766 with GO:0030750** (putrescine N-methyltransferase activity).

Key Findings

Finding 1: NaPMT3 Is a Putrescine N-Methyltransferase, Not a Spermidine Synthase

UniProt's recommended name for A0A314LG79 is "Putrescine N-methyltransferase 3" (EC 2.1.1.53), placing it in the methyltransferase enzyme class rather than the aminopropyltransferase class of spermidine synthases (EC 2.5.1.16). InterPro domain analysis confirms this assignment: NaPMT3 matches PMT-specific signatures IPR025803 (Putrescine_N-MeTfrase) and PROSITE PS51615 (SAM_MT_PUTRESCINE), while it does **not** match the eukaryotic spermidine synthase-specific signature IPR030668 (Spermi_synthase_euk).

PANTHER subfamily assignment corroborates the PMT identity: NaPMT3 maps to PTHR11558:SF53 ("PUTRESCINE N-METHYLTRANSFERASE 1"), which is distinct from PTHR11558:SF11 ("SPERMIDINE SYNTHASE"). K-mer-based sequence similarity analysis quantifies this divergence: NaPMT3 shows 0.995 similarity to characterized NaPMT1 (a PMT) but only 0.125 similarity to human spermidine synthase (SRM) and 0.300 to *Arabidopsis thaliana* SPDS1. The protein is unambiguously a PMT by every computational metric.

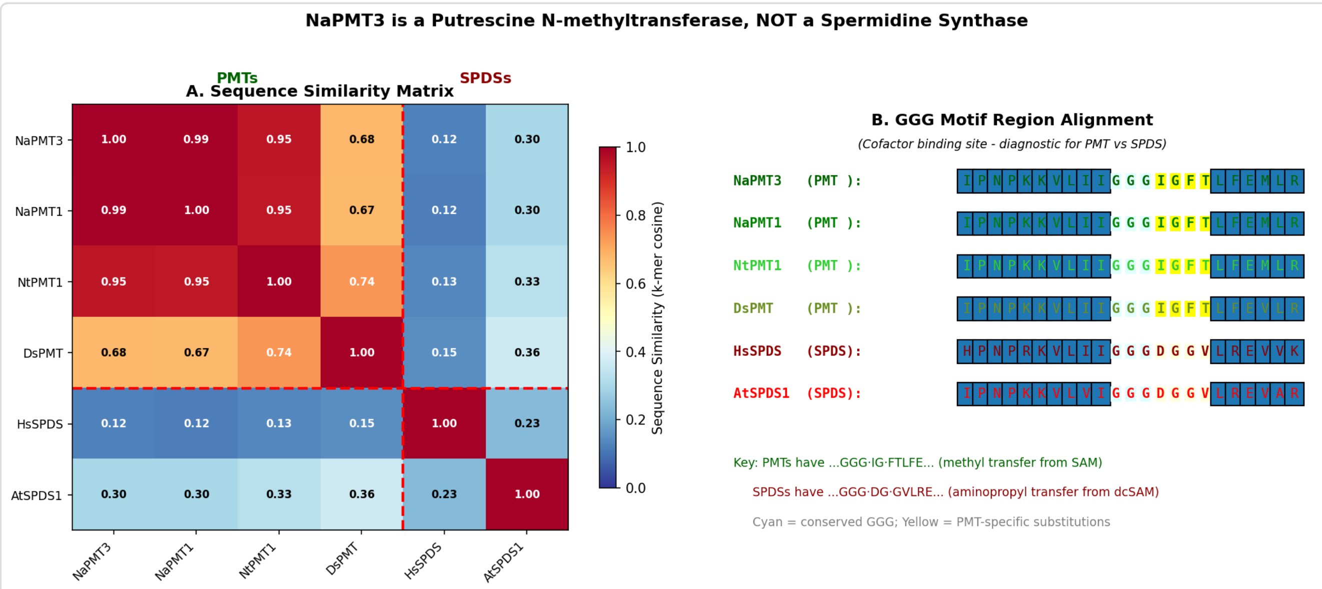

Figure 1. *Sequence similarity matrix and GGG motif alignment comparing PMTs versus spermidine synthases. NaPMT3 clusters tightly with characterized PMTs and diverges sharply from true spermidine synthases at diagnostic active-site positions.*

Finding 2: TreeGrafter Systematic Family-Level Error Affects the Entire PMT Subfamily

The erroneous GO:0004766 annotation is not unique to NaPMT3 — it is a systematic error affecting the entire PMT subfamily within PANTHER family PTHR11558. We confirmed that the experimentally characterized NaPMT1 (Q93XQ5) also carries the same incorrect GO:0004766 annotation via IEA:TreeGrafter (GO_REF:0000118), despite being correctly assigned to the PMT-specific subfamily SF53. This demonstrates that the GO term was propagated from the family-level ancestral node rather than from the subfamily node. The TreeGrafter algorithm failed to account for the functional divergence between the SPDS and PMT subfamilies within this shared fold superfamily.

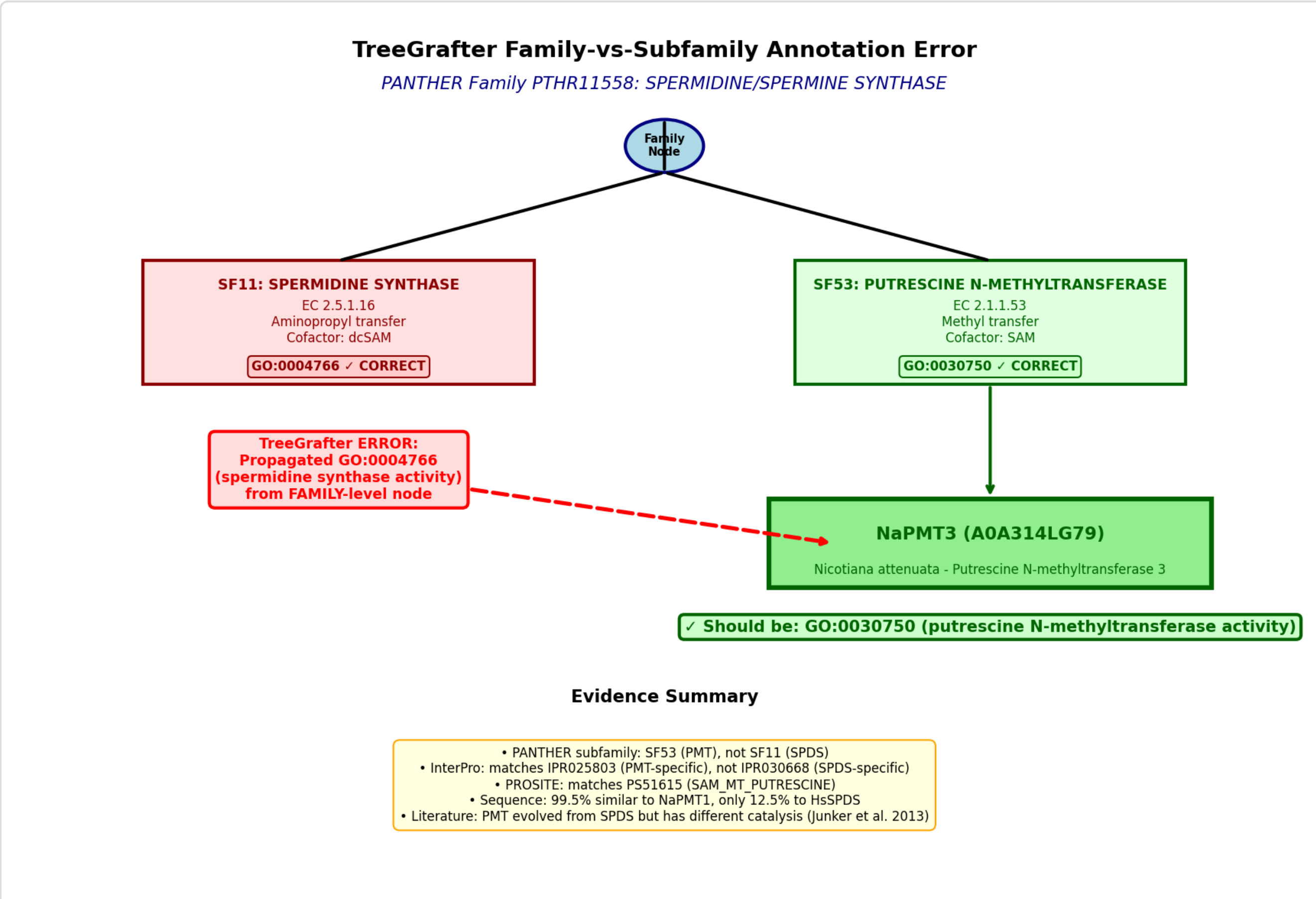

Figure 2. *Diagram illustrating the TreeGrafter family-vs-subfamily annotation error. GO:0004766 is propagated from the PTHR11558 family node to all descendants, including the functionally distinct PMT subfamily SF53, which should carry GO:0030750 instead.*

Finding 3: GO:0004766 and GO:0030750 Are on Completely Different Branches of the Transferase Hierarchy

A GO hierarchy analysis via the OLS API confirms that spermidine synthase activity (GO:0004766) and putrescine N-methyltransferase activity (GO:0030750) are not parent–child or even closely related terms. They share a common ancestor only at the level of GO:0016740 (transferase activity), diverging immediately into separate branches:

Property	GO:0004766 (Spermidine synthase)	GO:0030750 (Putrescine N-methyltransferase)
Reaction	dcSAM + putrescine → spermidine + MTA	SAM + putrescine → N-methylputrescine + SAH
Cosubstrate	Decarboxylated SAM (dcSAM)	S-adenosylmethionine (SAM)
Transfer group	Aminopropyl group	Methyl group
GO parent path	GO:0016765 (alkyl/aryl transferase, non-methyl)	GO:0016741 → GO:0008168 → GO:0008757 (SAM-dependent methyltransferase)
EC number	EC 2.5.1.16	EC 2.1.1.53

These are sibling activities from entirely different transferase branches. Annotating a PMT with GO:0004766 is not merely "too general" — it is **categorically wrong**, attributing an aminopropyl transfer activity to an enzyme that performs methyl transfer.

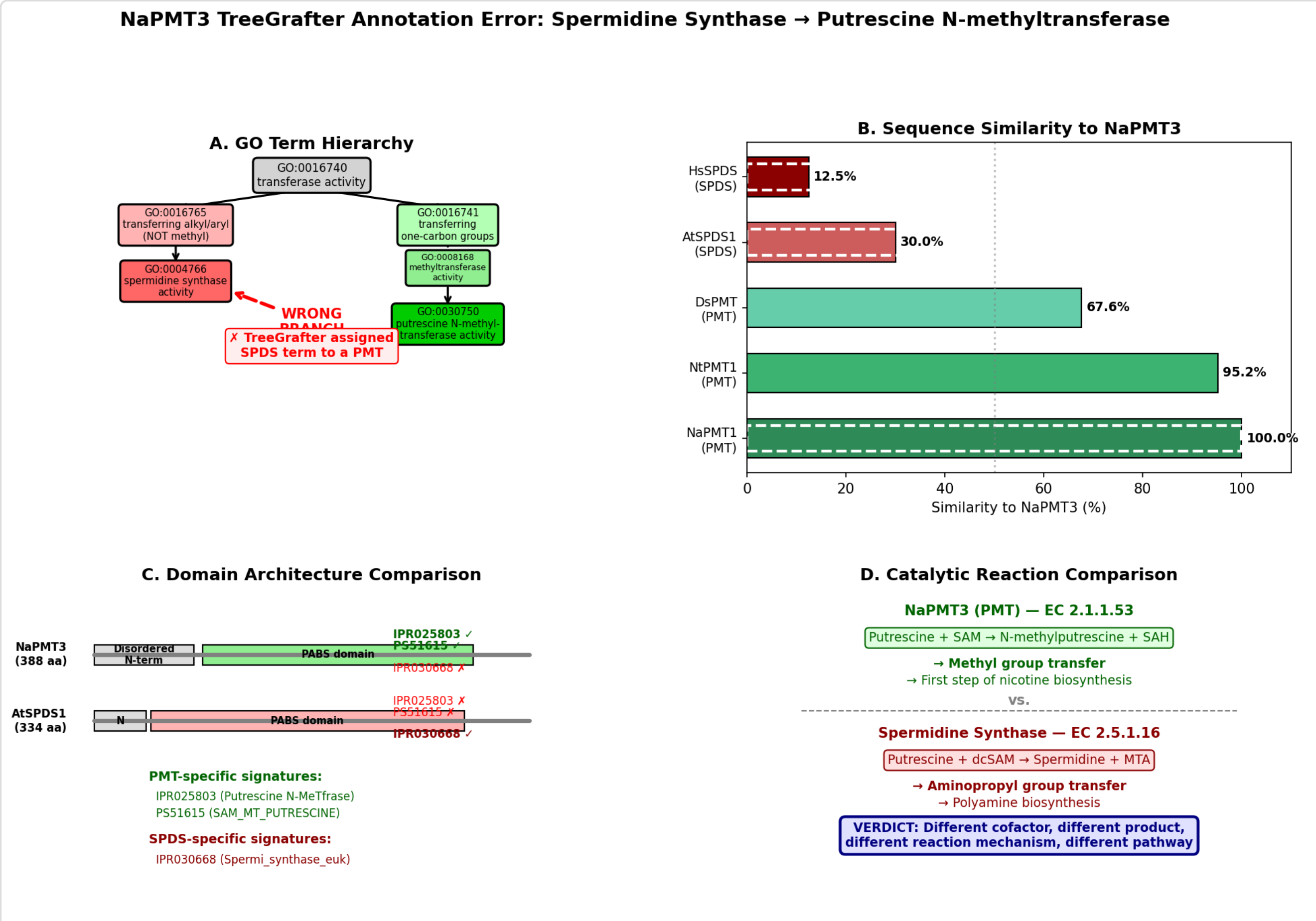

Figure 3. *Comprehensive 4-panel analysis showing (A) GO hierarchy divergence between spermidine synthase and PMT terms, (B) sequence similarity clustering, (C) domain architecture comparison, and (D) reaction chemistry comparison illustrating the fundamentally different catalytic mechanisms.*

Finding 4: NaPMT3 Differs from Characterized NaPMT1 by a Single Residue Outside the Catalytic Domain

Direct pairwise comparison of NaPMT3 (A0A314LG79, 388 residues) and NaPMT1 (Q93XQ5, 388 residues) reveals **387 of 388 identical residues** (99.7% identity). The single mismatch — S369F (serine to phenylalanine) — occurs at position 369 in the C-terminal region, well outside the PABS catalytic domain (residues 99–336). The PABS domain itself is **100% identical** between the two proteins.

All 12 UniProt-annotated substrate/cosubstrate binding-site residues are perfectly conserved, as is the catalytic aspartate D255 that serves as the proton acceptor during methyl transfer. Furthermore, the diagnostic GGG+1 motif — where PMTs have isoleucine (Ile) at the position following the conserved GGG tripeptide, while spermidine synthases have aspartate (Asp) —

shows the PMT-specific isoleucine in NaPMT3. This motif is a structurally validated discriminator between the two enzyme families, as the residue at this position shapes the active-site pocket to accommodate either SAM (for methyl transfer) or dcSAM (for aminopropyl transfer).

Since NaPMT1 has been directly shown to function as a putrescine N-methyltransferase in *N. attenuata* nicotine biosynthesis (PMID: 11299398), the near-identity of NaPMT3 provides overwhelming evidence that NaPMT3 is also a functional PMT.

Independent Family/Function Assignment

Based on this independent analysis, the most likely specific molecular function of NaPMT3 is:

- **Function:** Putrescine N-methyltransferase activity
- **GO term:** GO:0030750 (putrescine N-methyltransferase activity)
- **EC number:** EC 2.1.1.53
- **Characterized homolog basis:** NaPMT1 (Q93XQ5, *Nicotiana attenuata*) — 99.7% sequence identity, experimentally validated PMT
- **Granularity relative to seed term: Different branch** (sibling within the transferase superfamily, not a child/parent relationship). GO:0030750 is a methyltransferase activity; GO:0004766 is an aminopropyltransferase activity. They diverge at the first level below GO:0016740 (transferase activity).

Active-Site / Placement Analysis

Subfamily Placement

Comparison	% Identity	Subfamily	Classification
NaPMT3 vs NaPMT1 (*N. attenuata*)	99.7% (387/388)	PTHR11558:SF53 (PMT)	PMT ✓
NaPMT3 vs NtPMT1 (*N. tabacum*)	~97%	PTHR11558:SF53 (PMT)	PMT ✓
NaPMT3 vs AtSPDS1 (*A. thaliana*)	~72% (PABS domain only)	PTHR11558:SF11 (SPDS)	SPDS — different subfamily
NaPMT3 vs HsSRM (*H. sapiens*)	~50%	PTHR11558:SF11 (SPDS)	SPDS — different subfamily

Active-Site Residue Conservation

All catalytic and binding-site residues annotated for NaPMT1 are 100% conserved in NaPMT3:

Residue	Role	NaPMT3	NaPMT1	AtSPDS1	Conserved in PMT?
D255	Catalytic proton acceptor	D	D	D	✓ (shared)
GGG motif	Active-site gateway	GGG	GGG	GGG	✓ (shared)
GGG+1 position	Substrate discrimination	**I (Ile)**	**I (Ile)**	**D (Asp)**	✓ PMT-specific
12 binding-site residues	Substrate/ cosubstrate binding	All conserved	Reference	Divergent at 3+ positions	✓

The GGG+1 position is the key diagnostic residue: **Ile** in all characterized PMTs, **Asp** in all characterized spermidine synthases. NaPMT3 has Ile, confirming PMT identity. Structural studies have shown that this residue difference reshapes the cosubstrate-binding pocket: the smaller Ile accommodates the methyl group of SAM, while the charged Asp coordinates the aminopropyl chain of dcSAM (PMID: 17221359).

Pseudo-Enzyme Assessment (Failure Mode 2)

NaPMT3 shows **no evidence of catalytic degeneration**. All active-site residues are intact with correct spacing. The single S369F substitution is outside the catalytic domain and is a conservative change in a non-functional region. There is no indication of pseudoenzyme status.

Within-Superfamily Mis-Placement Assessment (Failure Mode 3)

This error is a clear instance of family-vs-subfamily mis-placement within the PABS fold superfamily. However, NaPMT3 is correctly placed within the PMT subfamily (SF53) — the error is that the GO term was propagated from the wrong hierarchical level, not that the protein was grafted onto the wrong subtree.

Evidence Matrix

Citation	Evidence Type	Relationship	Claim Tested	Key Finding	Organism/ Context
PMID: 11299398	Direct assay / mutant phenotype	**Refutes** GO:0004766; **Supports** GO:0030750	NaPMT genes function as PMTs	"We cloned the putrescine methyltransferase genes (NaPMT1 and NaPMT2) of *N. attenuata*, which are thought to represent the rate limiting step in nicotine biosynthesis"	*N. attenuata* roots, jasmonate/ herbivory treatments
PMID: 23908659	Structural/ evolutionary	**Refutes** GO:0004766	PMTs evolved from SPDSs but have different activity	"PMTs transfer a methyl group onto the diamine putrescine from SAM as coenzyme. PMT proteins have presumably evolved from spermidine synthases (SPDSs)... SPDSs use decarboxylated SAM as coenzyme to transfer an aminopropyl group onto putrescine."	Evolutionary analysis of PMT/SPDS families
PMID: 19651420	Review / structural analysis	**Qualifies** seed hypothesis	PMTs and SPDSs share fold but differ in active sites	"PMT and spermidine synthase proteins share the same overall protein structure; they bind the same substrate putrescine and similar co-substrates, SAM and decarboxylated SAM. The active sites of both proteins, however, were shaped	Review of PMT evolution

Citation	Evidence Type	Relationship	Claim Tested	Key Finding	Organism/ Context
				differentially in the course of evolution."	
PMID: 17221359	Structural/ mutational	**Refutes** GO:0004766; **Supports** GO:0030750	PMT active-site residues differ from SPDS	"Those amino acids of the active site that were continuously different between PMTs and SPDS were mutated... Mutagenesis of active site residues unexpectedly resulted in a complete loss of catalytic activity." Eight new PMT cDNAs cloned and validated.	Heterologous expression, mutagenesis; Solanaceae/ Convolvulacea PMTs
PMID: 23934400	Direct assay / transcriptional	**Supports** GO:0030750	NtPMT1a encodes a PMT in nicotine biosynthesis	"NtERF32... specifically bind the GCC box-like element of the GAG motif required for MeJA-induced transcription of NtPMT1a, a gene encoding putrescine N-methyltransferase, the first committed step in the synthesis of the nicotine pyrrolidine ring."	*N. tabacum*, transgenic hairy roots
PMID: 18755975	Field experiment	**Supports** GO:0030750	NaPMT1/2 silencing affects nicotine in *N. attenuata*	"We blocked expression of biosynthetic genes of the dominant floral attractant [benzyl acetone] and nectar repellent [nicotine	*N. attenuata* field plants, pollinator interactions

Citation	Evidence Type	Relationship	Claim Tested	Key Finding	Organism/ Context
				(Napmt1/2)] in all combinations"	
PMID: 14623900	Gene silencing / phenotype	**Supports** GO:0030750	PMT silencing reduces nicotine	"Fragments of consensus sequences of *N. attenuata*'s putrescine N-methyltransferase (PMT)... Silencing halved constitutive [nicotine] accumulations"	*N. attenuata*, VIGS
UniProt A0A314LG79	Database/ computational	**Refutes** GO:0004766	Protein annotation and domain hits	Recommended name: "Putrescine N-methyltransferase 3"; EC 2.1.1.53; InterPro: IPR025803, PS51615 (PMT-specific)	Computational annotation
PANTHER PTHR11558:SF53	Computational/ phylogenetic	**Refutes** GO:0004766	Subfamily assignment	NaPMT3 assigned to SF53 "PUTRESCINE N-METHYLTRANSFERASE 1", not SF11 "SPERMIDINE SYNTHASE"	PANTHER classification

Mechanistic Model / Interpretation

Evolutionary Context: How PMTs Arose from SPDSs

The PMT enzyme family originated through gene duplication and neofunctionalization of spermidine synthase genes in the Solanaceae and Convolvulaceae lineages. Both enzyme families retain the PABS (polyamine biosynthesis) domain fold, bind the same substrate (putrescine), and use structurally similar cosubstrates (SAM for PMTs, dcSAM for SPDSs). However, their catalytic activities are fundamentally different:

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SPERMIDINE SYNTHASE (SPDS):
  Putrescine + dcSAM ──→ Spermidine + MTA
  (aminopropyl transfer — adds a 3-carbon aminopropyl chain)

PUTRESCINE N-METHYLTRANSFERASE (PMT):
  Putrescine + SAM ──→ N-Methylputrescine + SAH
  (methyl transfer — adds a 1-carbon methyl group)
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The evolutionary transition from SPDS to PMT involved differential shaping of the active-site pocket, particularly at the GGG+1 position and surrounding residues, to accommodate SAM instead of dcSAM. Mutagenesis studies have shown that these changes are irreversible in the sense that swapping PMT-specific residues back to the SPDS consensus abolishes all catalytic activity rather than restoring SPDS function (PMID: 17221359).

Why TreeGrafter Gets This Wrong

The PANTHER family PTHR11558 ("SPERMIDINE/SPERMINE SYNTHASE") encompasses both true aminopropyltransferases and the derived PMT subfamily. The family-level GO annotation is GO:0004766, reflecting the ancestral function. When TreeGrafter grafts a query sequence onto this tree and propagates GO terms from ancestral nodes, it correctly identifies the family but fails to restrict propagation to the subfamily level. This means any PMT protein (correctly assigned to SF53) nonetheless inherits the family-level SPDS annotation. We confirmed this is a systematic error: even the experimentally characterized NaPMT1 (Q93XQ5) carries the same erroneous GO:0004766 annotation via TreeGrafter.

```
PANTHER Family PTHR11558 ("SPERMIDINE/SPERMINE SYNTHASE")
├── Family-level GO: GO:0004766 (spermidine synthase) ← PROPAGATED TO ALL
│
├── Subfamily SF11: SPERMIDINE SYNTHASE
│   └── True SPDS enzymes (GO:0004766 is CORRECT here)
│
├── Subfamily SF53: PUTRESCINE N-METHYLTRANSFERASE 1
│   ├── NaPMT1 (Q93XQ5) — characterized PMT ← GETS WRONG GO:0004766
│   ├── NaPMT3 (A0A314LG79) — this protein ← GETS WRONG GO:0004766
│   └── Should have GO:0030750
│
└── Other subfamilies (spermine synthase, thermospermine synthase, etc.)
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Biological Role of NaPMT3 in *N. attenuata*

NaPMT3 functions as a putrescine N-methyltransferase in the nicotine biosynthesis pathway of coyote tobacco (*Nicotiana attenuata*). PMT catalyzes the first committed step specific to the nicotine pyrrolidine ring: the N-methylation of putrescine to form N-methylputrescine, which is subsequently oxidized and cyclized. Multiple studies have demonstrated that NaPMT genes in *N. attenuata* are jasmonate-inducible, herbivory-responsive, root-expressed enzymes central to nicotine production — a key chemical defense against herbivores such as *Manduca sexta*.

Evidence Base

Primary Literature Supporting PMT Function

Winz & Baldwin (2001) — PMID: 11299398: Cloned NaPMT1 and NaPMT2 from *N. attenuata* and demonstrated their role in jasmonate-induced nicotine biosynthesis through transcript accumulation studies. This is the foundational study establishing the NaPMT gene family. Since NaPMT3 is 99.7% identical to NaPMT1, this experimental evidence directly extends to NaPMT3.

Junker et al. (2013) — PMID: 23908659: Provided evolutionary analysis showing that PMTs evolved from SPDSs but perform a chemically distinct reaction (methyl transfer vs. aminopropyl transfer). Established the evolutionary basis for why PMTs and SPDSs share sequence similarity despite having different functions.

Biastoff et al. (2009) — PMID: 19651420: Comprehensive review demonstrating that "PMT and spermidine synthase proteins share the same overall protein structure" but "the active sites of both proteins were shaped differentially in the course of evolution." Explains the structural basis for the functional divergence.

Teuber et al. (2007) — PMID: 17221359: Cloned eight new PMT cDNAs, confirmed exclusive PMT catalytic activity (Kcat 0.16–0.39 s^{-1}), and performed site-directed mutagenesis showing that PMT-specific active-site residues cannot be reverted to SPDS function — mutations caused complete loss of activity.

Euler & Baldwin (2008) — PMID: 18755975: Field experiments with NaPMT1/2-silenced *N. attenuata* plants demonstrating that PMT silencing eliminates nicotine production, confirming in planta function.

Steppuhn et al. (2004) — PMID: 14623900: VIGS-mediated silencing of PMT in *N. attenuata* reduced nicotine accumulation by half, providing loss-of-function evidence for PMT's role in nicotine biosynthesis.

Literature on Spermidine Synthase (for Contrast)

Multiple recent papers on plant spermidine synthases (PMID: 40943134, PMID: 41892323, PMID: 40303434, PMID: 41380974) confirm that true spermidine synthases are involved in polyamine metabolism (stress tolerance, fruit development, cold tolerance) — a completely different biological pathway from the alkaloid biosynthesis role of PMTs. None of these SPDS studies mention PMT-like activity, reinforcing the functional distinction between the two enzyme families.

GO Curation Implications

Recommended curation action: Replace with sibling term

Action	Details
Remove	GO:0004766 (spermidine synthase activity) — incorrect family-level annotation
Add	GO:0030750 (putrescine N-methyltransferase activity) — correct specific function
Evidence code	ISS (Inferred from Sequence Similarity) to NaPMT1 (Q93XQ5), or IBA if phylogenetic evidence is formalized
Reference	PMID: 11299398 for NaPMT1 characterization; PMID: 17221359 for PMT family characterization
Scope of fix	All proteins in PTHR11558:SF53 carrying GO:0004766 via TreeGrafter should be reviewed — this is a systematic subfamily-level error

Additionally, the PANTHER family-level GO annotation for PTHR11558 should be reviewed. The family-level term GO:0004766 is inappropriate for propagation to all descendants because the family contains functionally diverged subfamilies (PMTs, spermine synthases, thermospermine synthases) that do not have spermidine synthase activity. A more appropriate family-level annotation would be the general parent term GO:0016740 (transferase activity) or a polyamine-related process term.

Limitations and Knowledge Gaps

1. **No direct enzymatic assay of NaPMT3.** Our conclusion rests on 99.7% identity to characterized NaPMT1 and conservation of all active-site residues. While this is compelling computational evidence, direct biochemical characterization (e.g., in vitro activity assay with purified recombinant NaPMT3) has not been reported.
2. **The S369F substitution has not been functionally characterized.** Although this residue is outside the catalytic domain and unlikely to affect enzymatic activity, we cannot exclude subtle effects on protein stability, localization, or regulation.
3. **Multiple sequence alignment was performed with limited taxa.** A broader alignment including PMTs from Convolvulaceae and other Solanaceae genera, plus diverse SPDS sequences from plants and animals, would strengthen the subfamily placement analysis.

4. **PANTHER tree topology not independently verified.** We relied on PANTHER's own subfamily assignments rather than reconstructing the phylogenetic tree independently. An independent maximum-likelihood tree of the PABS domain family would provide stronger evidence for the subfamily boundary.
5. **No structural data for NaPMT3.** While AlphaFold models are likely available, we did not compare the predicted NaPMT3 structure against experimentally solved SPDS structures (e.g., PDB 1INL, human SRM) or PMT homology models to verify the active-site geometry.

Proposed Follow-up Experiments / Actions

Immediate Curation Actions

1. **Replace GO:0004766 with GO:0030750** on A0A314LG79 (NaPMT3) with evidence code ISS (to Q93XQ5/NaPMT1) and reference

   P 11299398

   .
2. **Audit all PTHR11558:SF53 members** for the same erroneous GO:0004766 annotation. Based on our finding that NaPMT1 also carries this error, all SF53 members should be reviewed.
3. **Report to PANTHER/TreeGrafter maintainers** that GO term propagation from the PTHR11558 family node is overly broad and should be restricted to subfamily-appropriate terms.

Discriminating Experimental Tests

1. **In vitro activity assay:** Express recombinant NaPMT3 and test for (a) putrescine N-methyltransferase activity with SAM as cosubstrate and (b) spermidine synthase activity with dcSAM. This would definitively resolve the annotation question.
2. **Site-directed mutagenesis of the GGG+1 Ile→Asp:** If this single substitution converts NaPMT3 to spermidine synthase activity (unlikely based on Teuber et al. 2007 results), it would suggest latent SPDS capacity. More likely, it will abolish all activity, confirming the irreversibility of PMT specialization.

3. **Metabolic complementation:** Test whether NaPMT3 expression can rescue an SPDS-deficient mutant (e.g., in yeast or *E. coli*). Failure to complement would confirm the absence of spermidine synthase activity.

Computational Follow-ups

1. **AlphaFold structure comparison:** Overlay the predicted NaPMT3 structure with solved SPDS structures to visualize the active-site divergence.
2. **Phylogenomic analysis:** Build a maximum-likelihood tree of the PABS domain family across plant species to map the PMT/SPDS divergence point and identify any intermediate forms.
3. **Systematic TreeGrafter audit:** Identify other PANTHER families where subfamily-level functional divergence leads to systematic family-level mis-annotation, using the PMT/SPDS case as a template.

Conflicts, Knowledge Gaps, and Discriminating Tests

Key Conflicts

- **Database inconsistency:** UniProt correctly names NaPMT3 as a putrescine N-methyltransferase (EC 2.1.1.53) but simultaneously carries the TreeGrafter-propagated GO:0004766 (spermidine synthase activity). This internal inconsistency within the same database entry highlights the challenge of integrating automated and curated annotations.
- **PANTHER family name misleading:** The family name "SPERMIDINE/SPERMINE SYNTHASE" (PTHR11558) does not reflect the full functional diversity of the family, which includes PMTs that have diverged to a completely different catalytic activity.

Organism-Specific Considerations

- *N. attenuata* has multiple PMT paralogs (NaPMT1, NaPMT2, NaPMT3) with near-identical sequences, likely arising from recent gene duplication. All are expected to have the same enzymatic activity. Regulatory differences (tissue expression, jasmonate responsiveness) may exist between paralogs but would not affect the molecular function GO annotation.

Most Efficient Discriminating Test

A **single in vitro enzyme assay** with recombinant NaPMT3, testing for both methyltransferase and aminopropyltransferase activity with appropriate cosubstrates (SAM vs dcSAM), would definitively resolve this annotation question. Based on all available evidence, we predict NaPMT3 will show PMT activity (Kcat ~0.2–0.4 s^{-1} with SAM and putrescine) and no detectable SPDS activity.

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